1. Role of JDK, JRE and JVM

JDK – Java Development Kit

- JDK is used by **developers to create Java applications**.
- It contains tools required to **write, compile, debug, and run** Java programs.
- The important tool is **javac**, which converts Java source code (.java) into bytecode (.class).
- Example: A developer uses JDK to create and compile a Java program.

JRE – Java Runtime Environment

- JRE provides the **environment required to run Java applications**.
- It includes the **JVM and required Java runtime libraries**.
- It is mainly concerned with **running** Java programs rather than developing them.
- Example: A user needs a Java runtime environment to execute a compiled Java application.

JVM – Java Virtual Machine

- JVM is the **engine that executes Java bytecode**.
- It loads and verifies the .class files and executes the bytecode.
- JVM is **platform-dependent**, meaning different operating systems have their own JVM implementations.
- Example: Windows, Linux, and macOS use JVM implementations suitable for their respective platforms.

Simple Relationship

JDK → JRE → JVM

- **JDK** = Develop Java programs
- **JRE** = Run Java programs
- **JVM** = Execute Java bytecode

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2. How Java Achieves Platform Independence

Java achieves platform independence through **bytecode and the JVM**.

1. The programmer writes Java source code in a .java file.
2. The **Java compiler (javac)** converts the source code into **bytecode** (.class).
3. This bytecode is not directly dependent on a particular operating system.
4. A suitable **JVM** on each operating system executes the same bytecode.
5. Therefore, the same Java program can run on **Windows, Linux, macOS**, etc., without changing the source code.

Example

Java Source Code

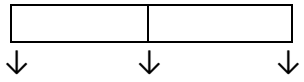
↓

javac

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Java Bytecode (.class)

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JVM JVM JVM

Windows Linux macOS

This is why Java is commonly described as:

“Write Once, Run Anywhere (WORA)”

Java Architecture — JDK, JRE and JVM

How Java source code becomes a platform-independent application

Java Source Code

Example: HelloWorld.java

javac compiler

JDK — Java Development Kit

Development tools:

- javac — Java compiler
- java — application launcher
- javadoc — documentation tool
- Debugging, packaging and other development utilities

compiled bytecode

JRE — Java Runtime Environment

Provides the environment required to run Java applications

JVM — Java Virtual Machine

Loads, verifies and executes Java bytecode

machine-level execution

Operating System & Hardware

Windows • Linux • macOS • Other supported platforms

The JVM provides the platform-specific implementation needed to run the bytecode.

Platform Independence

Java source → bytecode → JVM → platform-specific execution

The same .class bytecode can run on different operating systems when a compatible JVM is available.